Backend Behind All This

IIEST Student-Teacher AI Portal

A simple technical explanation of how the IIEST Student-Teacher AI Portal works behind the scenes, written so it can be explained to teachers, evaluators, or project reviewers.

Core Tech Used

- Frontend: Streamlit is used for the web interface. All pages like Teacher Portal, Live Exam, BerriBot, Grading, Papers, Dashboard, and Learning Resources are Streamlit pages.

- Backend: FastAPI is used as the API server. The frontend sends requests to backend endpoints for login, exam creation, AI tutoring, grading, question bank access, enrollments, results, and analytics.

- Database: SQLite with SQLAlchemy stores users, exams, questions, student enrollments, teacher subjects, results, assignments, and exam set allocations.

- Authentication: JWT token authentication is used. After login, the backend returns a token, and protected pages/API calls use that token to verify whether the user is a student or teacher.

- AI: AI features go through a centralized /ai_tutor API. The system can use local AI through Ollama/Llama if available, and fallback academic logic keeps the portal functional if the AI server is unavailable.

- Voice: BerriBot uses the browser Web Speech APIs. SpeechRecognition listens to the student, and SpeechSynthesis speaks the AI answer aloud.

- File Reading/OCR: Uploaded PDFs, DOCX, text files, CSV/Excel files, and images are passed through extraction logic. Clear text documents are read directly; scanned or unclear handwritten files may require teacher correction.

1. Login And Role Access

Students and teachers log in through the same portal. The backend checks credentials and sends back a JWT token. The frontend stores this token and uses it whenever it calls a protected API.

Teachers can access teacher tools and student preview tools because they need to inspect the learner experience. Students cannot access teacher-only pages.

If asked how it works: login -> backend verifies user -> backend returns token -> frontend stores token -> every page and API checks the role before allowing access.

2. Subject Enrollment System

Students enroll themselves into subjects from the Student Dashboard by entering name, roll number, semester, section, subject, and subject code.

Teachers enroll themselves for the subjects they teach from the Teacher Portal.

The backend matches teacher subject code, semester, and section with student enrollments. This lets the teacher see the exact roster for each subject.

Why it matters: exams, analytics, attendance, grading, and remedial support become subject-wise instead of random.

3. Teacher Portal

The Teacher Portal is the command center. It includes daily priorities, class capture, subject roster, remedial cohorts, AI question studio, syllabus progression, and LMS-style tools.

AI helps teachers by identifying students needing attention, suggesting teacher actions, generating tests from taught topics, preparing remedial groups, and supporting grading.

The standout idea is teacher intelligence: not just storing data, but helping the teacher decide what to do next.

4. Class Capture

A teacher can record or type what was taught in class. The portal converts that lecture summary into notes and a short concept-check test.

Flow: teacher selects subject and subject code -> starts capture or types summary -> AI summarizes taught topics -> teacher creates an after-class check -> notes can be sent to enrolled students.

How to explain it: instead of preparing notes and tests separately after class, the teacher gives the lecture content once, and the system turns it into usable student material.

5. Exam Builder

Teachers create exams by choosing subject, subject code, topics, question type, difficulty, number of questions, number of sets, duration, and deadline.

The backend filters the structured question bank by subject, topic, difficulty, and question format. It then builds Set A, Set B, Set C, etc. automatically.

The teacher does not manually write each question or assign each paper. The system creates balanced sets from the question bank.

6. Question Bank

The question bank is structured for B.Tech CSE-style subjects and supports MCQs, short answers, long answers, numerical/problem-solving questions, and coding-style questions where applicable.

Each question stores metadata such as subject, topic, difficulty, type, marks, options, correct answer, keywords, and explanation.

This avoids the old placeholder problem where every MCQ looked like Correct Answer / Wrong Answer. Questions are generated and selected based on subject and topic.

7. Live Exam

Students see teacher-created exams in the Live Exam page. Exams show subject, subject code, duration, deadline, and status.

Flow: teacher creates exam -> backend saves exam/questions -> student opens Live Exam -> starts exam -> browser timer runs -> student submits -> backend stores results.

MCQs can be auto-marked. Short/long/coding answers are stored for manual or AI-assisted evaluation.

8. Exam Set Assignment

When a teacher creates multiple sets, the portal automatically divides enrolled students alphabetically and as evenly as possible across Set A, Set B, Set C, etc.

The backend stores which student received which set, so the teacher has a record later.

Why it matters: this supports realistic exams where different sets reduce copying but teachers still need clear tracking.

9. AI Grading Center

Typed assignment grading allows the teacher to paste or type a student answer and a rubric. AI evaluates concept coverage, keywords, missing points, and feedback.

Uploaded paper grading lets the teacher upload a PDF/DOC/image. The system extracts text where possible. If handwriting is unclear, the teacher can correct the extracted text before grading.

This is realistic because OCR is not perfect for handwritten copies. The system supports AI grading, but keeps the teacher in control.

10. Attendance Intelligence

Teachers can maintain attendance records and use AI to identify risk patterns.

AI does not just show percentage. It highlights low attendance, repeated absence, possible risk, and suggested teacher action.

This makes attendance useful for intervention, not just record keeping.

11. Analytics Dashboard

The dashboard reads exam results, submissions, attendance, and subject data from the database.

AI helps interpret this data by identifying weak topics, students needing support, difficult questions, and revision needs.

Teachers can use it to understand class progress and change records where required.

12. Remedial Cohorts

The system groups students who are weak in similar topics or subjects.

Example: students weak in DBMS normalization can be grouped separately from students weak in operating system scheduling.

The AI checks results, weak-topic data, and performance signals to suggest groups, so teachers do not manually inspect every paper.

13. Student Dashboard

Students can enroll in subjects, view exams, track results, access weak-subject revision, and use AI study tools.

The dashboard is connected to the student login, subject enrollment, exam attempts, and result records.

14. BerriBot / TeachMeBerriBot

BerriBot is the AI voice study assistant for students.

Universal knowledge mode answers from general academic knowledge. Uploaded material first mode uses the uploaded document first, then general knowledge. Uploaded material only mode answers only from the uploaded document.

Voice flow: student clicks Listen -> browser records speech -> transcript goes to /ai_tutor -> backend generates answer -> browser speaks the answer using speech synthesis.

Voice commands like pause, stop, resume, and continue are handled in the browser.

Why it stands out: students can learn through live spoken interaction instead of only typing into a chatbot.

15. Papers Section

Students can upload previous year papers, PDFs, notes, and question papers.

The system extracts text and lets students ask AI to summarize, explain, or create a revision plan from the uploaded paper.

Example use case: upload a DBMS question paper and ask for a chapter-wise revision plan.

16. Learning Resources

Students enter a topic and receive structured suggestions such as YouTube learning links, study material ideas, articles, notes, PDFs, and reference resources.

The AI generates a guided learning path for deeper understanding.

17. Weak Subjects / Improvement Generator

Students select weak subjects or topics. The system creates a revision plan, daily practice reminders, practice questions, and improvement strategy.

This feature connects self-assessment with AI-generated academic planning.

Best Way To Present The Project

Say this: This is not just an exam website. It is an AI-supported academic ERP/LMS for teachers and students.

The teacher side reduces manual work like test creation, grading, attendance tracking, class capture, remedial grouping, and syllabus planning.

The student side supports AI tutoring, voice learning, exam practice, weak-topic revision, paper analysis, and personalized study support.

Technical summary: The frontend is Streamlit, the backend is FastAPI, the database is SQLite with SQLAlchemy, authentication uses JWT, and AI features go through the /ai_tutor API. Voice uses browser speech recognition and speech synthesis.